

ITEC614 - Cyber Security Concepts

Assessment 2: Reflective Report on JB Hi-Fi Alleged Data Breach

Semester 1, 2025

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Article words count: 873

Appendix I: Feed Forward

Questions	Feedback recommendation
How have you acted on the feedback from the previous assignment to improve your work in this assignment?	Based on my past criticism, I made beyond any doubt that I backed my talk, position, thoughts, and ideas with peer-reviewed journal references in this task.
What is your expectation around the type of feedback that enhances your learning?	I need to know where I made a botch and how I can rectify them and not make the same botch once more i.e., I need criticism that will help me to move forward with my learning and execution within another task.
Did you have any difficulty understanding or acting on previous feedback? Please be as specific as possible so that you can gain further feedback/clarify anything you do not understand in the feedback	The input given in my past task was exceptionally bland; I did not know how to improve my work. So, I would just like the instructor to clarify more on related subject viewpoints of the input, or I would like an opportunity to have an exchange to get it the input.

Appendix II: Assignment Checklist

1. My submitted assignment report is within the specified word limit	✓
2. I have included references using the specified referencing style	✓
3. I have correctly cited all my sources and references	✓
4. I have formatted my report as per the specifications	✓
5. I have checked my Turnitin report to ensure the similarity report is within the acceptable range level (below 20% similarity)	✓
6. I have received feedback from the previous assignment.	✓
7. I have completed proofreading and checked for spelling and grammar	✓
8. I have submitted my work before the due date/time	✓
9. I have submitted the feed-forward template along with my assignment submission	✓

Contents

Appendix I: Feed Forward..... I

Appendix II: Assignment Checklist..... II

1. Introduction 1

2. Details of Breach Causes 1

 i. Critical Factors Leading to the Security Breach Claim 1

 ii. Tactics and Techniques Used by Hackers 1

 iii. Company's Identification and Remediation Efforts 2

 iv. Existing Security Measures 2

 v. Sufficiency and Currency of Security Measures 2

3. Critical Security Controls..... 2

Conclusion..... 2

References 2

Appendix III: Assessment 2 DF Engagement..... III

JB Hi-Fi Alleged Data Breach

A.Tamang, P.P.Pradhan

Abstract—In January 2025, an individual using the alias "LordAbe" asserted on a hacking forum that they had bought and were offering for sale more than 12 million customer details allegedly from JB Hi-Fi, which is an Australian electronics retailer. The dataset contained data like names, email addresses, mobile numbers, and postcodes. Yet, further questions showed the sample data supplied were in line with records of a past security breach for Dymocks, which is also an organization from Australia. JB Hi-Fi immediately denied any allegations of unauthorized intrusion into their systems, indicating the information was not in line with their internal records and that they do not keep certain personal information, including identity documents or financial details. This incident underscores the difficulty for organizations in substantiating claims of violations and emphasizes the need for holistic cybersecurity practices and response measures in a timely manner to forestall reputational loss due to unfounded claims.

Index Terms—security, attacks, defenses, breach

1. Introduction

The cyber incident on JB Hi-Fi in early 2025 represents a significant shift in cyber-attacks, where false information is becoming the primary mechanism to strike. Unlike traditional data breaches based on technical vulnerabilities, this incident displays how malicious entities can employ fear and skepticism to destabilize a firm's reputation. The misleading claim of hacking into 12 million customer records that turned out to be untrue brought severe challenges to the firm and briefly dissolved stakeholders'

confidence. This report examines the incident from three fronts: technical analysis of the potential attack vector, evaluation of the company's crisis management, and examination of broader industry implications. (Troy Hunt, 2025) The analysis could not be timelier, with similar reputation-oriented attacks increasing globally, more recently against financial participants and healthcare organizations. By studying JB Hi-Fi's example, we derive valuable insights that most organizations across industries can implement to manage comparable threats to their reputation in the wake of an increasingly digitally driven business environment.

2. Details of Breach Causes

i. Critical Factors Leading to the Security Breach Claim

The primary factor leading to the breach claim was the threat actor's assertion of possessing a vast dataset allegedly from JB Hi-Fi. However, analysis revealed that the sample data provided matched information from a prior breach involving Dymocks, indicating that the claim was unfounded and based on recycled data. (Apple Podcasts, 2025)

ii. Tactics and Techniques Used by Hackers

Applying the MITRE ATT&CK framework, the threat actor's tactics align with:

- T1588.001 – Obtain Capabilities: Malware: Acquiring previously leaked data to feign a new breach.
- T1585.001 – Establish Accounts: Social Media Accounts: Using online forums to disseminate false claims.

- T1591.002 – Gather Victim Identity Information: Employee Names: Leveraging known data to fabricate breach narratives.

iii. Company's Identification and Remediation Efforts

JB Hi-Fi promptly addressed the claim, stating that they were unaware of any unauthorized data extraction and that the sample data did not correspond with their records. They emphasized that their systems do not store certain personal information, such as identity documents or financial details, further discrediting the breach claim. (Richards, 2025)

iv. Existing Security Measures

While specific security measures were not detailed publicly, JB Hi-Fi's swift response and ability to verify the integrity of their data suggest the presence of robust cybersecurity protocols, including data monitoring and incident response strategies.

v. Sufficiency and Currency of Security Measures

The company's immediate and transparent handling of the situation indicates that their security measures are both sufficient and up to date, enabling them to effectively counter false breach claims and maintain stakeholder trust.

3. Critical Security Controls

To protect against such kinds of attacks, organizations need to have the security controls below in place:

- Monitoring and Logging: Use monitoring sensors to detect attempts to access your information or suspicious activity in real time.
- 4 Classification and Minimization of Data Minimize the amount of stored user information and classify it

according to its sensitivity for the application of protection measures.

- At least, a plan relating to incident response: Prepare and maintain a response plan for both non-issues and those that begin to surface as common breaches.
- Public Comms Protocols: Put in place clear comms so you can quickly and transparently counter public breach claims.
- Third-Party Risk Management: Assess and monitor third-party vendors often, verifying their compliance with rigorous cyber security rules to minimize the likelihood of having data exposed by an external partner. (Richards, 2025)

Conclusion

The false data breach claim against JB Hi-Fi in January 2025 demonstrates the challenges faced by organizations in the current day-and-age of misinformation, when damaging reputational impact can happen extremely quickly. JB Hi-Fi provides a positive example of incident management being conducted quickly and openly, underlining the importance of both solid cyber security protections and communication plans. Amid today's ever-changing cyber threat landscape, organizations need to stay on their toes, with frameworks that are flexible and robust enough to fend off valid attacks and false assertions.

References

Apple Podcasts, 2025. *Apple Podcasts*. [Online] Available at: <https://podcasts.apple.com/us/podcast/24-01-2025-jb-hi-fi-data-claims-hpe-breach-investigation/id1792148329?i=1000685218687> [Accessed 10 May 2025].

Attack Mitre, 2025. *Attack Mitre*. [Online]
Available at: <https://attack.mitre.org/>
[Accessed 10 May 2025].

Croft, D., 2025. *Cyber Daily.AU*. [Online]
Available at:
<https://www.cyberdaily.au/security/11623-exclusive-threat-actor-falsely-claims-leak-of-12m-jb-hi-fi-records>
[Accessed 10 May 2025].

Richards, D., 2025. *Channel News*. [Online]
Available at:
<https://www.channelnews.com.au/breaking-news/jb-hi-fi-denies-massive-cyber-attack-claim>
[Accessed 10 May 2025].

Troy Hunt, 2025. *Troy Hunt*. [Online]
Available at: <https://www.troyhunt.com/you-cant-trust-hackers-and-other-data-breach-verification-tales/>
[Accessed 10 May 2025].

Appendix III: Assessment 2 DF Engagement

Piyush Prasad Pradhan:



Piyush Prasad Pradhan



4 Apr 15:03

In recent years, the rapid growth in the use of Internet of Things (IoT) devices has significantly enhanced business operations but also has introduced new types of cybersecurity risks. In 2017, a North American casino suffered a major data breach when attackers exploited a vulnerable smart fish tank thermometer to access organizations secured internal network. This breach in the system enabled cybercriminals to exfiltrate 10GB of confidential data, including credentials and confidential details of high value customer. Primary factors contributing to this attack included weak IoT security controls and updates of security patches, lack of network segmentation and monitoring. Attackers used tactics like privilege escalation, lateral movement and data exfiltration as outlined in the MITRE framework. This incident highlights urgent need for organizations to address IoT-related security risks to prevent unauthorized access of organizations critical system.

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Piyush Prasad Pradhan



4 Apr 15:03

To defend against such dangerous cyber-attacks, businesses organizations must implement essential security controls like network segmentation, strong authentication and continuous monitoring. Isolating IoT devices on separate VLANs can prevent lateral movement, while enforcing multi-factor authentication (MFA) and limited access ensures that unauthorized user access cannot exploit weak security. Performing regular firmware and patch updates are crucial in terminating known vulnerabilities. Furthermore, organizations should use intrusion detection and prevention systems (IDPS) and AI-driven threat analytics to identify and neutralize potential threats. As well as, encrypting all sensitive data and using secure communication protocols to maximize overall cybersecurity defenses.

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Piyush Prasad Pradhan



4 Apr 15:08

The breach of a casino in 2017 is a powerful reminder that IoT ecosystems need to be properly protected. A proactive approach, incorporating regular assessments of risk, penetration testing and stringent access control policies, can help to mitigate the risks of smart devices. These initiatives aim to provide enterprises with a greater sense of security. Data security should not be a risk that new IoT products create or run. Business organizations need to employ a Zero Trust security model, invest in cyber security first, and always be vigilant against the latest cyber threats that are emerging from an increasingly networked world.

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Piyush Prasad Pradhan



11 Apr 22:06

In conclusion, this report examines how a smart fish tank thermometer was used in a 2017 casino cyberattack. The breach was caused due to weak security on the IoT devices, poor network design and a lack of monitoring. The attackers used methods described in the MITRE ATT&CK framework to move through the network and steal sensitive data of the organization. To prevent similar attacks, companies should use strong passwords, keep devices and patches updated, use separate networks and use tools to watch for threats. In the future, more IoT devices will be used in businesses, so it's mandatory and important to have simple but strong security plans to prevent cyber-attacks. Use of AI to detect problems beforehand and keeping up with security rules as well as teaching employee about safe practices can help protect systems. With above steps organizations and businesses can reduce the risk of cyberattacks and make their networks safer.

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Piyush Prasad Pradhan

10 May 18:35

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Amir Tamang:

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Assignments

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2025 Semester 1

Week 6 Tasks:

This week, you will be writing the conclusion and abstract part of your report. Please follow the assessment 2 guide when writing conclusion and abstract. Your conclusion section must include a summary of the report with future directions. On the other hand, the abstract should highlight the significance of the report and highlight key findings of your research.

Week 7 Tasks:

This week, you need to compile week 3 to week 6 posts in a single report. Use the IEEE template to format your report and acknowledge all academic resources you used to write the report using IEEE referencing style. **Upload the word file as an attachment.**

Reply

AT **Amir Tamang**

10 May 15:40

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